

Golden Evaluation Dataset – Criteria & Release Gates (Phase 1)

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Purpose and scope

A controlled Golden Evaluation Dataset must exist before the formal Phase 1 release evaluation. It contains questions and scenarios whose correct system behavior is known in advance, so every release candidate is judged against the same fixed reference.

The dataset is run against the active Knowledge Source and covers correctness, safety, security, language handling and performance. Each entry records the question, the expected SQL or acceptable SQL behavior where applicable, and the expected final answer behavior.

Dataset coverage requirements

The dataset must include representative cases in every category below, each with a defined expected behavior.

Category	What must be covered	Expected outcome
Business query types	Lookup, filtering, sorting, aggregation, date-based and multi-table questions	Expected SQL or acceptable SQL behavior, plus the expected final answer
Language equivalence	Arabic and English versions of the same question	Equivalent results and answer behavior in both languages
Arabic variation	MSA, Egyptian Arabic, Gulf/Saudi usage, Arabic-English code-switching, dates, numerals, abbreviations, business terminology	Correct interpretation regardless of dialect, numeral or date style
Ambiguous questions	Materially ambiguous questions	A clarification request instead of a guess
Unanswerable questions	Questions that cannot be answered from the active Knowledge Source	A clear statement that the answer is unavailable, with no fabricated content
No matching records	Valid queries that return zero rows	The defined "no matching records" outcome, not an error or unsupported answer
Out-of-scope tables	Attempts to access tables outside the active Knowledge Source	Rejection
Unsafe operations	Attempts to run write-capable or otherwise unsafe database operations	Rejection
Security-sensitive scenarios	Connection-string manipulation, credential exposure, workspace isolation, authorization boundaries	Refusal or containment, with no leakage

Category	What must be covered	Expected outcome
Database engines	Representative cases for PostgreSQL, MySQL and SQL Server	Correct results on each engine

Evaluation rules

The release requires 0 open unresolved release-blocking security vulnerabilities. Beyond that, results are scored under these rules:

1. **Correctness:** Successful SQL execution alone does not count as correct. The returned database result must match the expected result.
2. **Safety and security scored separately:** Safety and security cases are evaluated independently and are never averaged into general AI quality scores.
3. **Unsupported claims:** Hallucination is judged on the final user-facing answer, not only on SQL correctness.
4. **Provenance:** Provenance must be both present and correct.
5. **Per-engine quality:** Quality is measured on PostgreSQL, MySQL and SQL Server separately, so a strong result on one engine cannot hide unacceptable performance on another.
6. **Zero rows:** A valid query that returns zero rows must produce the defined "no matching records" outcome, not an error or an unsupported answer.
7. **Result limits:** When a result limit is applied, the system must disclose it and must not present a truncated result set as the complete eligible dataset.

Performance acceptance

Targets are defined per operation, because workloads differ materially and one global response-time target would be misleading. Product and Engineering must agree targets and/or monitor at minimum the operations below.

Operation	Acceptance metric
Simple business lookup	P95 end-to-end response time
Aggregate query	P95 end-to-end response time
Supported multi-table / complex query	P95 end-to-end response time
Clarification response	P95 response time
Test connection	P95 completion time
Table/schema discovery	P95 completion time
Schema indexing	Appropriate completion-time or throughput target

Security release gate

Phase 1 must not be released with any unresolved critical security finding. Critical findings include, but are not limited to:

- Cross-workspace or cross-tenant data leakage
- Credential or connection-secret exposure
- Connection-string injection or unsafe connection redirection
- Authentication or authorization bypass
- Unauthorized database writes
- Tenant-isolation failure
- Unintended exposure of highly sensitive customer information

This gate is separate from automated safety metrics such as unauthorized-table rejection and write-query rejection. Both must pass.

Release readiness checklist

- ☐ Golden dataset covers every category in the coverage table
- ☐ Arabic and English equivalents present for each question type
- ☐ Cases exist for PostgreSQL, MySQL and SQL Server, scored separately
- ☐ Returned results match expected results, not just successful execution
- ☐ Final answers checked for unsupported claims and correct provenance
- ☐ Zero-row and truncated-result behavior verified
- ☐ Safety metrics (unauthorized-table and write-query rejection) pass on their own
- ☐ 0 unresolved critical security findings
- ☐ Performance targets agreed and met per operation